

BUSINESS REPORT

Hypertension Control & Treatment Effectiveness Analysis

Using SQL Server and Microsoft Excel

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Tools Used: SQL Server Management Studio (SSMS), Microsoft Excel

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1. Executive Summary

Hypertension is one of the leading risk factors for cardiovascular diseases and requires continuous monitoring to ensure effective treatment outcomes. Healthcare providers rely on patient data to evaluate whether prescribed medications are successfully controlling blood pressure and to identify opportunities for improving patient care.

This project analyzed a dataset containing 350 hypertension patients to evaluate blood pressure control after three months of treatment. SQL Server was used for data exploration, validation, and analytical querying, while Microsoft Excel was used to create calculated fields, perform pivot table analysis, and develop an interactive dashboard.

The analysis focused on treatment effectiveness, medication adherence, and blood pressure control outcomes. The results showed that only a small proportion of patients achieved controlled blood pressure, while patients with good medication adherence demonstrated better treatment outcomes. Combination Therapy recorded the highest blood pressure control rate among the treatment options evaluated.

The interactive dashboard provides healthcare stakeholders with an accessible tool for monitoring treatment outcomes and supporting data-driven clinical decision-making.

2. Introduction

Hypertension remains a significant public health challenge and is one of the primary contributors to heart disease, stroke, and kidney failure. Effective management depends not only on selecting appropriate medications but also on ensuring that patients adhere to their prescribed treatment plans.

Healthcare organizations routinely collect patient data, yet much of this information remains underutilized. Transforming clinical data into meaningful insights enables healthcare professionals to identify treatment gaps, evaluate medication effectiveness, and make informed decisions that improve patient outcomes.

This project demonstrates how SQL Server and Microsoft Excel can be combined to analyze hypertension treatment data and present actionable insights through an interactive dashboard.

3. Project Objectives

The primary objectives of this project were to:

1. Assess blood pressure control after three months of treatment.
2. Evaluate the relationship between medication adherence and treatment outcomes.
3. Compare the effectiveness of different antihypertensive drug classes.
4. Develop an interactive dashboard to support healthcare decision-making.

5. Present analytical findings in a clear and visually engaging format.

4. Dataset Overview

The analysis was conducted using a dataset containing 350 patient records.

The dataset includes demographic information, blood pressure measurements, medication adherence status, prescribed antihypertensive drug classes, and treatment outcomes after three months.

■ Key Variables

Category	Variables
Patient Information	Patient ID, Age, Gender
Clinical Measurements	Baseline Systolic BP, Baseline Diastolic BP, Follow-up Systolic BP, Follow-up Diastolic BP
Treatment Information	Antihypertensive Drug Class
Patient Behaviour	Medication Adherence
Outcome	BP Controlled After 3 Months

5. Methodology

The project followed a structured analytics workflow:

■ Step 1 – Data Import

The hypertension dataset was imported into SQL Server Management Studio for analysis.

■ Step 2 – Data Validation

SQL queries were used to:

- Verify the total number of patient records.
- Check for duplicate Patient IDs.
- Identify missing values in key variables.

■ Step 3 – Exploratory Data Analysis

SQL queries were used to:

- Analyze blood pressure control outcomes.
- Evaluate medication adherence.

- Compare treatment effectiveness across drug classes.

■ Step 4 – Dashboard Development

The validated dataset was exported to Microsoft Excel where:

- PivotTables were created.
- Calculated fields (Age Group and Diastolic Reduction) were added.
- PivotCharts were developed.
- KPI cards and slicers were designed.
- An interactive dashboard was created.

6. Data Quality Assessment

Before conducting the analysis, the dataset was assessed for quality to ensure the reliability of the results.

The validation process confirmed that:

- ✓ The dataset contained 350 patient records.
- ✓ No duplicate Patient IDs were identified.
- ✓ No missing values were detected in the key variables used for analysis.

The dataset was considered complete and suitable for exploratory analysis and dashboard development.

7. Analysis and Findings

The analysis produced several important insights:

■ Blood Pressure Control

Only 9.43% of patients achieved controlled blood pressure after three months of treatment, while 90.57% remained uncontrolled. This suggests that a significant proportion of patients may require additional interventions or treatment adjustments.

■ Medication Adherence

Patients with good medication adherence demonstrated the highest blood pressure control rate. Patients with poor adherence showed the lowest treatment success, highlighting the importance of consistent medication use.

■ Treatment Effectiveness

Among the antihypertensive drug classes analyzed, Combination Therapy achieved the highest blood pressure control rate (21.05%), outperforming the other treatment options included in the dataset.

■ Age Group Analysis

Blood pressure control remained relatively low across all age groups, indicating that hypertension management remains a challenge regardless of patient age.

8. Dashboard Overview

The final dashboard was developed in Microsoft Excel to provide an interactive summary of the analysis.

The dashboard includes:

- Total Patients KPI
- Blood Pressure Control Rate KPI
- Non-Controlled Rate KPI
- Top Performing Drug Class KPI
- Medication Adherence Analysis
- Drug Class Effectiveness Analysis
- Average Diastolic Reduction
- Blood Pressure Control by Age Group
- Interactive slicers for filtering results

Dashboard Snapshot

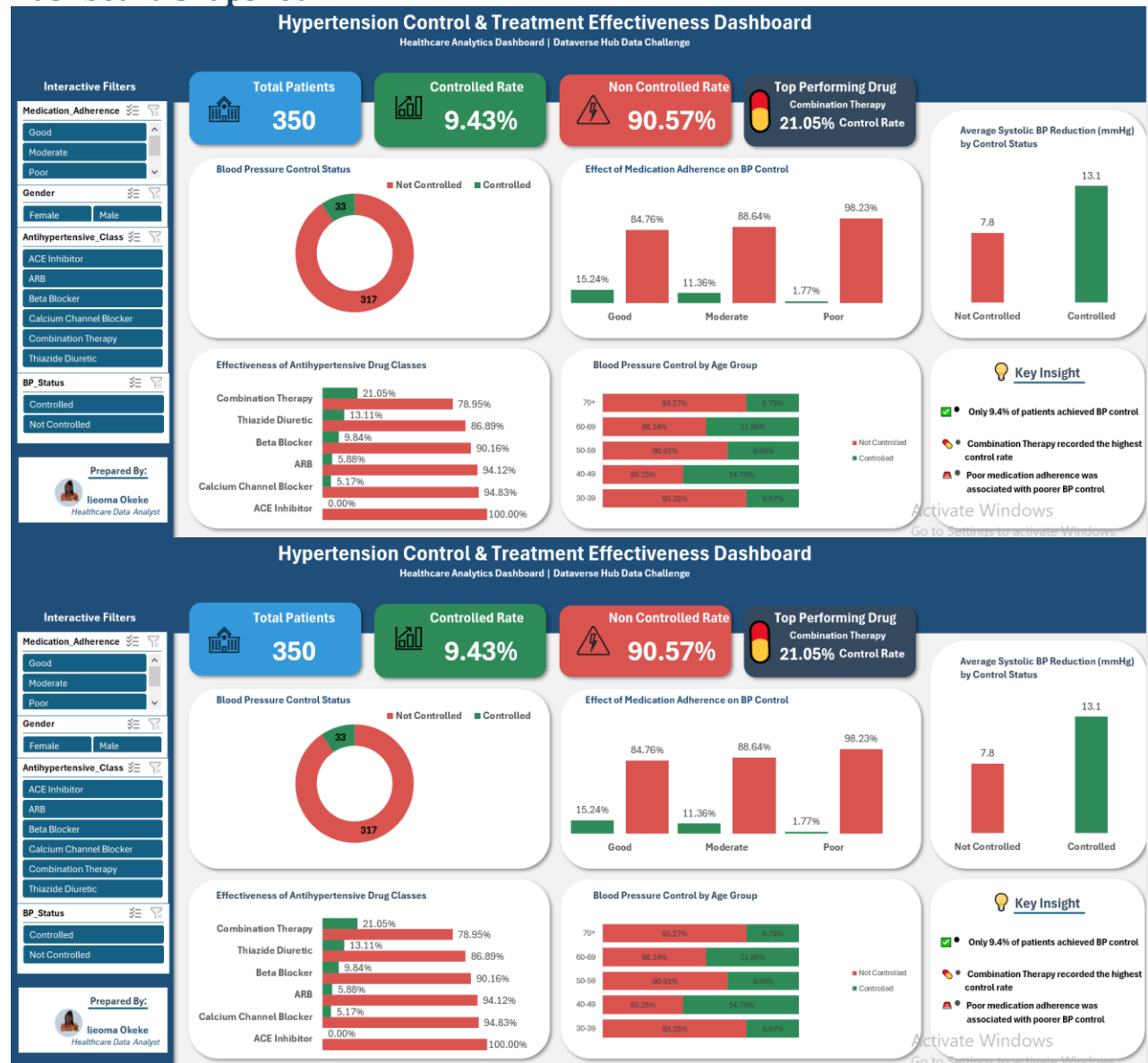


Figure 1: Hypertension Control & Treatment Effectiveness Dashboard

9. Recommendations

Based on the findings, the following recommendations are proposed:

1. Strengthen patient education programs to improve medication adherence.
2. Increase follow-up monitoring for patients with uncontrolled blood pressure.
3. Evaluate the suitability of Combination Therapy for eligible patients based on clinical guidelines.
4. Use interactive dashboards to support continuous monitoring of hypertension treatment outcomes.
5. Encourage data-driven decision-making to improve patient care and treatment effectiveness.

10. Conclusion

This project demonstrates the value of combining SQL Server and Microsoft Excel to transform healthcare data into actionable insights.

By applying structured data validation, exploratory analysis, and interactive dashboard design, the project provides a practical framework for monitoring hypertension treatment outcomes. The findings highlight opportunities to improve medication adherence and optimize treatment strategies, while the dashboard offers a user-friendly tool for supporting evidence-based healthcare decisions.

Author's Note

This report is designed to accompany the SQL script, Excel dashboard, and presentation as part of a complete healthcare analytics portfolio piece.